

Assignment 2

Please submit the code and report (in pdf format) on Blackboard system before 23:59 April 26. Report can be written either in English or Chinese. Name your report as `studentID_Name.pdf` . The report contains the screenshot of the program result (display first 5 lines). You can include the explanation of the key functions/codes in the report or in the code as comments.

Q1

The data of `departuredelays.csv` is attached. Complete the following task **using Spark RDD API**.

(1) [2 point] Partition the data according to the `origin` columns. Put the rows where `origin` is `ATL` into one partition, while the rest rows are randomly partitioned into other three partitions.

Q2

Here is a dataset related to the online open courses. The `courses.csv` contains the information about the courses while `instructors.csv` contains the information about the instructors.

(1) [1 points] Join the two dataframes by `instructors_id`, use the `inner_join` function.

(2) [2 points] Use SQL (in Pyspark SQL) to show the `display_name` and `job_title` of the instructor, who has the highest course rating among all courses that are related to 'spark' (i.e. contains the word 'spark') and are created after 2018-01-01 00:00:00.

(3) [2 points] Use SQL (in Pyspark SQL) to select all courses that are (a) related to 'interview' (i.e. contains the word 'interview' or 'interviews') and then (b) sorted by `course_rating` in descending order and `created` in descending order (newest first). In this task, the course rating should be firstly rounded to one decimal place (e.g., 4.67748 -> 4.7).

Q3

The database of `debit_card_specializing` is attached in file `Q3_data/debit_card_specializing/debit_card_specializing.sqlite`. Write code to answer the following question, "What is the difference in the annual average consumption of the customers with the least amount of consumption paid in CZK for 2013 between SME and LAM, LAM and KAM, and KAM and SME?"

The evidence to this question is "annual average consumption of customer with the lowest consumption in each segment = total consumption per year / the number of customer with lowest consumption in each segment; Difference in annual average = SME's annual average - LAM's annual average; Difference in annual average = LAM's annual average - KAM's annual average; Year 2013 can be presented as Between 201301 And 201312; The first 4 strings of the Date values in the yearmonth table can represent year."

(1) [1 point] write code with Spark dataframe API to answer this question.

(2) [2 points] write code with Spark SQL API to answer this question.

You are expected to use **LLM/agents** for this task. You can build some easy toy data for the verification of generated code. The attached `hw02_Q3.ipynb` can be used to import sqlite database into pyspark with `sqlite-jdbc-3.51.3.0.jar`.